

The Frontline Exposure Gap: Evidence on AI Adoption in Retail and Service Occupations from Task-Level Usage Data

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Abstract

This paper measures realized AI adoption in frontline service occupations using the Anthropic Economic Index, which maps millions of AI conversations to O*NET task statements. Frontline occupations in sales, office and administrative support, food preparation and serving, and personal care and service account for 31.7 percent of U.S. employment but only 11.1 percent of task-matched AI usage. Three findings characterize this gap. First, occupational classification understates it: excluding technical occupations classified under clerical codes reduces the administrative-support representation index from 0.65 to 0.34, whereas a task-splitting robustness check leaves the estimates unchanged. Second, comparisons of the February and August 2025 releases provide suggestive evidence that the gap persisted rather than narrowed. Third, conditional on reaching frontline tasks, AI usage is more likely to involve automation-style interactions. In frontline administrative-support tasks, 62.7 percent of classified conversations are automation-style, compared with 43.0 percent in other occupations. Experimental research associates this interaction pattern with weaker skill formation. Occupation-level regressions show that the gap is not primarily a wage gradient. The estimated wage elasticity of usage is 0.38 (robust SE = 0.19), and the coefficient on a frontline indicator is statistically indistinguishable from zero. These results are consistent with a deficit that operates through lower usage per worker rather than lower usage per occupation. February 2026 exposure scores further reveal polarization within frontline work between screen-mediated and physically co-present service occupations. I discuss demand, access, and measurement mechanisms and their distributional implications.

1. Introduction

While operating Krishna Foods and Energy, I observed a sharp divide between the tasks performed in the back office and those performed on the production and retail floor. Demand forecasts, supplier comparisons, pricing analyses, customer correspondence, and management reports could be digitized and increasingly supported by AI. Packing, stocking, serving customers, supervising production, and resolving real-time quality problems remained physical, time-sensitive, and difficult to mediate through a chat interface. The task-level data analyzed in this paper reproduce that practical divide: AI appears first in the information tasks embedded within frontline occupations, while the in-person core of the work remains largely absent. This observation motivates the paper's central question.

This paper asks where AI adoption actually stands in frontline service work, which employs nearly one-third of American workers, and what characterizes the adoption that does occur.

The question bears directly on the distribution of gains from generative AI. Where the technology has been deployed, estimated productivity effects are large and often largest for less-experienced workers.

Brynjolfsson, Li, and Raymond (2025) document a 14 percent average increase in issues resolved per hour among customer-support agents given an AI assistant, and Noy and Zhang (2023) find comparable gains in professional writing tasks. These estimates are conditional on deployment. If adoption systematically bypasses occupations in which many low- and middle-wage workers are employed, equalization within a task can coexist with a widening divide between occupations. Adoption inequality is documented directly in Danish linked survey-register data: Humlum and Vestergaard (2025) find chatbot take-up concentrated among younger, higher-earning workers, with women 16 percentage points less likely to adopt.

The paper's measurement contribution is to observe adoption rather than predict it. Prospective exposure indices score occupations by whether their tasks could, in principle, be performed by AI (Felten, Raj, and Seamans 2021; Eloundou et al. 2024). The Anthropic Economic Index (AEI; Handa et al. 2025) instead classifies millions of actual Claude conversations to O*NET task statements. This distinction is substantive: frontline occupations score as moderately exposed in prospective indices but are only weakly represented in observed usage. When they do appear, the interaction modes also differ from those emphasized in much of the productivity literature.

I document five results. First, frontline occupation groups hold 31.7 percent of employment and 11.1 percent of task-matched usage; representation indices range from 0.64 for administrative support to 0.06 for food service, compared with 10.9 for computer and mathematical occupations. Second, a taxonomy audit shows that the administrative-support estimate is inflated by technical occupations historically classified under clerical codes. Third, the February and August 2025 releases show little movement toward parity, although differences in their classification pipelines prevent a formal trend estimate. Fourth, conditional on observed use, frontline tasks are disproportionately associated with automation-style interactions. Fifth, by February 2026, observed exposure had polarized within frontline work: screen-mediated service occupations ranked among the most exposed in the economy, while physically co-present service occupations remained close to zero.

The analysis is descriptive and makes no causal claims. Its empirical discipline lies in measurement validity. Each headline estimate is subjected to robustness exercises involving taxonomy correction, task splitting, cell-size disclosure, and platform-selection analysis. These checks are designed to establish whether the reported patterns are features of observed adoption rather than artifacts of classification or sample composition.

2. Conceptual Framework

A task-based framework organizes the empirical analysis (Acemoglu and Autor 2011; Acemoglu and Restrepo 2019). An occupation is a bundle of tasks, and AI enters at the task level. For a task to appear in usage data, three conditions must hold jointly: technical feasibility, worker access to an appropriate device and interface at the point of work, and sufficient expected value to the user. A fourth condition operates at the level of the firm: realizing value from AI typically requires complementary organizational investments, including workflow redesign and changes to decision structures, so falling technology costs alone need not produce adoption (Agrawal, Gans, and Goldfarb 2018). Prospective exposure indices measure primarily the first condition; observed usage reflects all of them. The wedge between exposure and usage is therefore informative about access, value, and organizational frictions, although it cannot identify them separately.

Define the representation index for occupation group g as

$$R_g = s_g^U / s_g^E \quad (1)$$

where the numerator is group g 's share of task-matched usage and the denominator is its share of employment. A value of 1 indicates usage proportional to employment. Values below 1 can arise through two channels: fewer occupations in the group appearing in usage at all, or usage failing to scale with the number of workers in each occupation. Section 5.3 examines usage intensity among occupations with positive observed usage. Because the regression sample excludes zero-usage occupations, it does not fully identify the extensive margin.

The framework also distinguishes interaction modes. Following Handa et al. (2025), directive and feedback-loop conversations delegate the task and are classified as automation-style, while iteration, learning, and validation conversations work through the task and are classified as augmentation-style. In the displacement-reinstatement language of Acemoglu and Restrepo (2019), automation-style usage substitutes for worker task performance, whereas augmentation-style usage complements it. Experimental evidence also links full delegation to weaker skill formation (Shen and Tamkin 2026). The mode composition of frontline usage is therefore an outcome of interest in its own right.

3. Data

Usage. The initial AEI release, published in February 2025, reports the share of a large sample of Claude.ai conversations classified to each of 3,514 *ONET task statements by Clio, a privacy-preserving aggregation pipeline* (Tamkin et al. 2024; Handa et al. 2025). The V3 release, published in September 2025, reports analogous shares for an August 4-11, 2025 window across 2,617 matched tasks and classifies each task's conversations into five collaboration modes. After text normalization, the task statements match ONET-SOC occupation codes exactly at the statement level. This match rate governs the data join, not the accuracy of the upstream conversation-to-task classifier.

Employment and wages. Employment counts are drawn from the BLS Occupational Employment and Wage Statistics for May 2023 and aggregated to SOC major groups. Occupation-level median wages are drawn from O*NET.

Observed exposure, February 2026. The AEI labor-market-impacts release reports an observed-exposure score for 756 occupations (Massenkoff and McCrory 2026). The score combines O*NET task data, task-level feasibility ratings (Eloundou et al. 2024), and observed usage through February 2026. It measures the employment-relevant share of an occupation's tasks with observed AI coverage. Because this is a coverage measure rather than a usage-share measure, it is treated as a complementary cross-section rather than as a direct continuation of the 2025 series.

Definitions and conventions. Frontline occupations are SOC major groups 41 (Sales and Related), 43 (Office and Administrative Support), 35 (Food Preparation and Serving Related), and 39 (Personal Care and Service). Usage shares are renormalized over task-matched conversations, excluding the V3 'none' bucket of 5.7 percent. Employment shares are fixed at May 2023 values, so changes in the representation index reflect changes in measured usage rather than changes in employment.

Known limitations. The data capture Claude usage rather than all AI usage. User occupation is inferred from task content rather than observed directly, and conversation share is not equivalent to the share of work time. Employer-deployed systems may also be underrepresented in consumer conversation logs. Section 6 assesses these limitations and the direction in which each might affect the results.

4. Empirical Approach

The design is descriptive. The estimands are population shares, ratios of shares, and conditional means in observed usage data. No counterfactual is constructed and no causal parameter is identified. Instead of a conventional identification strategy, the analysis addresses four threats to measurement validity: occupational misclassification within the SOC taxonomy, many-to-many task-occupation mappings, small cells in mode-composition estimates, and platform selection into Claude usage. Sections 5 and 6 address these threats using an exclusion audit, a fractional-assignment exercise, cell-size disclosure, and a platform-selection assessment supported by within-platform contrasts and external survey evidence. Causal-adjacent language such as 'enters' or 'absorbs' refers only to observed patterns.

For the occupation-level analysis, I estimate by ordinary least squares

$$\ln(s_o^U) = a + b \ln(w_o) + c \text{Frontline}_o + e_o \quad (2)$$

with heteroskedasticity-robust HC1 standard errors. The sample contains occupations with positive observed usage and wage data. The model therefore describes variation in usage intensity among observed occupations; it does not estimate the probability that an occupation appears in the usage data.

5. Results

5.1 Descriptive Statistics

Frontline groups jointly hold 31.7 percent of U.S. employment and 11.1 percent of task-matched usage. The representation indices are 0.64 for office and administrative support, 0.26 for sales, 0.23 for personal care and service, and 0.06 for food preparation and serving. Computer and mathematical occupations have an index of 10.9, accounting for 37.3 percent of usage but only 3.4 percent of employment. In per-worker terms, retail sales work appears in the usage data at approximately one-fortieth of the rate of software work, and food service at approximately one-hundred-eightieth of the rate.

Table 1. Representation indices by occupation group

Occupation group	Raw index	After taxonomy correction
Office and administrative support	0.64	0.34
Sales and related	0.26	0.26
Personal care and service	0.23	0.23
Food preparation and serving	0.06	0.06
Computer and mathematical	10.94	10.94

Note: The corrected administrative-support estimate excludes four technical occupations classified under SOC 43. Other groups are unchanged apart from rounding.

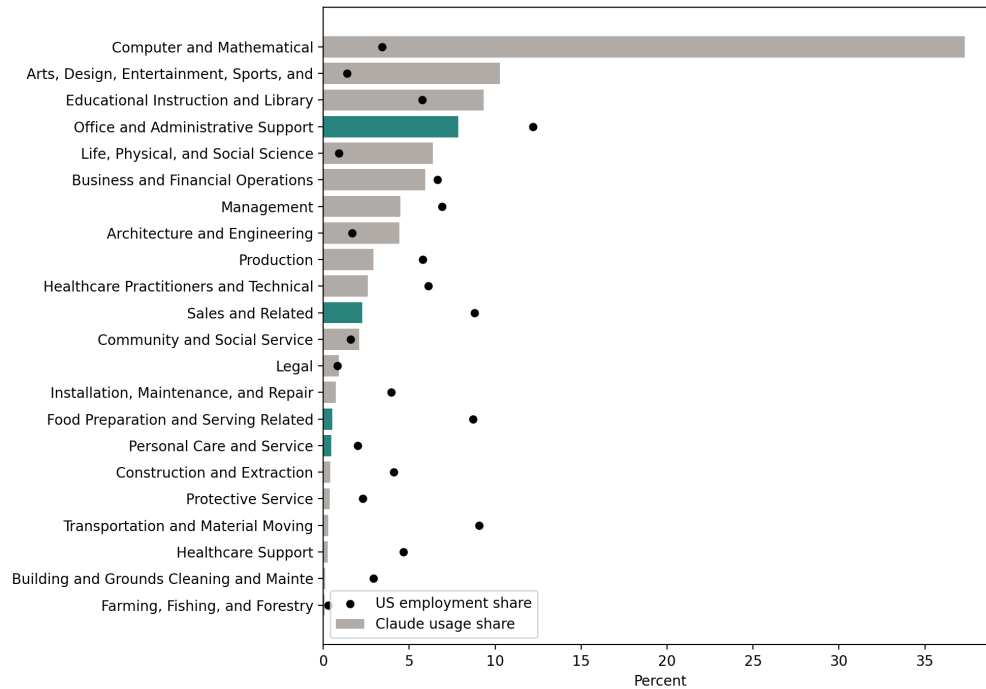


Figure 1. AI usage and employment shares across SOC major groups.

Notes: Bars show each group's share of task-matched Claude conversations; points show its share of U.S. employment. Frontline groups are highlighted in teal. Sources: Anthropic Economic Index and BLS Occupational Employment and Wage Statistics, May 2023.

Within the raw frontline classification, the highest-usage tasks are primarily information tasks. Several of the leading entries, however, are technical tasks attached to occupations historically filed under administrative codes. Figure 2 displays the raw task ranking to make this classification problem visible. After removing the misclassified technical occupations, the leading genuinely frontline tasks include composing responses to customer correspondence, answering procedural and policy questions, planning events and services, reformatting documents, and supervisory problem resolution. The in-person core of the occupations (selling, serving, and caring) is largely absent.



Figure 2. Highest-usage tasks in the raw frontline SOC classification.

Notes: The figure intentionally reports the raw ranking. Technical tasks near the top motivate the taxonomy audit in Section 5.2. Long O*NET task statements are truncated. The corrected interpretation is based on the occupation exclusions described in Section 5.2.

5.2 Taxonomy Audit

The administrative-support estimate is inflated by the occupational classification system. Its highest-usage tasks are disproportionately attached to technical occupations filed under SOC 43's historical clerical categories: Bioinformatics Technicians (43-9111.01), Computer Operators (43-9011), Statistical Assistants (43-9111), and Desktop Publishers (43-9031). Excluding these four occupations reduces the administrative-support representation index from 0.645 to 0.338. The remaining frontline indices move by no more than 0.01, indicating a targeted rather than mechanical correction. The corrected estimates place all four frontline groups between 0.06 and 0.34.

5.3 Wages and the Margin of the Gap

Using the 585 occupations with positive observed usage and wage data, the estimated wage elasticity is 0.38 (SE = 0.19) without the frontline indicator and 0.41 (SE = 0.21) with it. The R-squared is 0.03 in both specifications, so wages explain little of the cross-occupation variation in usage. The frontline coefficient is 0.19 (SE = 0.19). The estimate is statistically indistinguishable from zero and is positive, although imprecisely estimated.

Table 2. Occupation-level usage regressions

Variable	(1)	(2)
Log median wage	0.38 (0.19)	0.41 (0.21)
Frontline occupation		0.19 (0.19)
Observations	585	585

R-squared | 0.03 | 0.03

Notes: The dependent variable is log occupation usage share. HC1 robust standard errors are in parentheses. The sample includes occupations with positive observed usage and wage data.

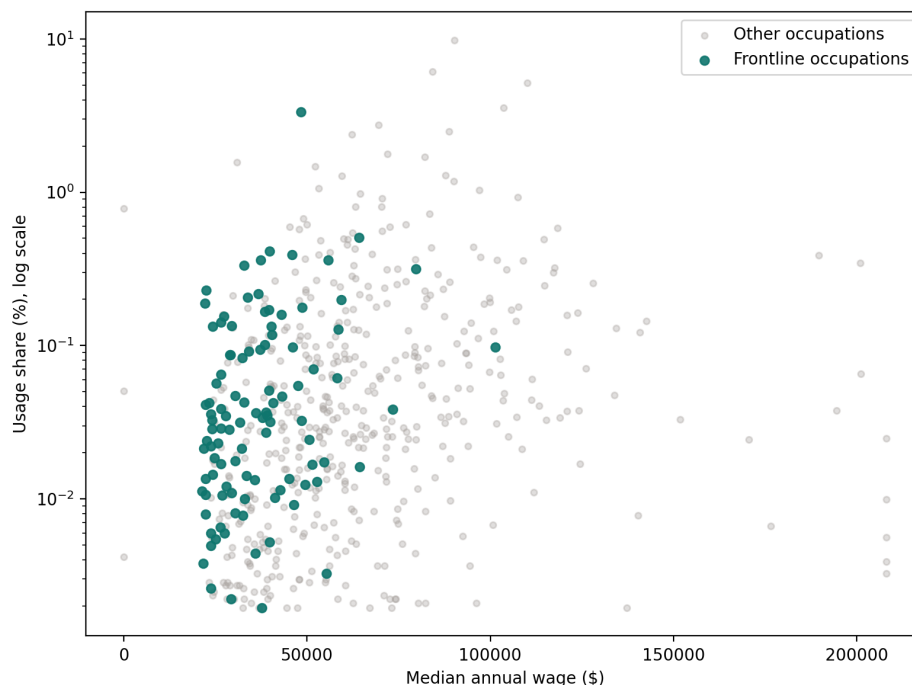


Figure 3. Occupation-level usage share and median annual wage.

Notes: The vertical axis is logarithmic. Teal points denote frontline occupations; gray points denote other occupations. The plotted scale is descriptive and differs from the log-log specification reported in Table 2. N = 585 occupations with positive usage and wage data.

These regressions describe usage intensity among occupations that appear in the usage data. They do not fully separate the extensive and intensive margins because zero-usage occupations are excluded. Within the positive-usage sample, wages explain little and frontline status adds no negative association. Combined with the large employment counts in frontline occupations, this pattern is consistent with usage failing to scale with the number of workers in each occupation. A two-part model that separately estimates entry into observed usage and conditional usage intensity remains an important extension.

5.4 Dynamics, February-August 2025

Across the two releases, representation indices moved modestly: computer and mathematical occupations increased from 10.94 to 11.48, office and administrative support from 0.65 to 0.73, sales from 0.26 to 0.30, and personal care and service from 0.23 to 0.34; food preparation and serving declined from 0.061 to 0.045. No frontline group approached parity, and the overall gap remained large. Because the releases use different classification pipelines and task universes (3,514 tasks in the initial release and 2,617 in V3), the levels are not strictly commensurable. I therefore interpret the comparison as suggestive evidence of continued underrepresentation rather than as a measured trend. A methodologically consistent panel remains a future extension rather than a completed one. The June 2026 AEI report, *Cadences* (Massenkoff et al. 2026), introduces continuous hourly and daily telemetry

and a linked user survey, but as of this writing its underlying data files are not part of the public repository, and its published description emphasizes usage rhythms and survey responses rather than confirming an occupational task-usage panel comparable to the two 2025 releases used here. Whether it can support the panel outlined in Section 8 should be verified against the actual data files, if and when released, rather than assumed.

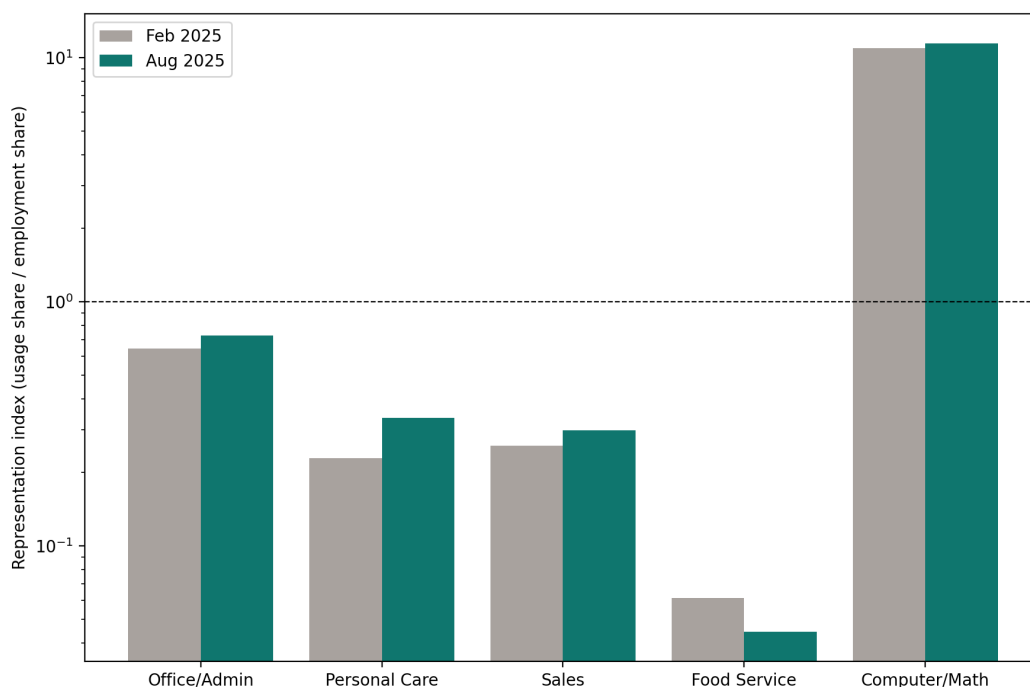


Figure 4. Representation indices in the February and August 2025 AEI releases.

Notes: The dashed line marks parity between usage share and employment share. The horizontal axis is logarithmic. The two releases use different classification pipelines and task universes, so the comparison is descriptive rather than a formal panel estimate.

5.5 Interaction Modes

Conditional on reaching frontline tasks, usage is more strongly tilted toward delegation. Automation-style shares are 62.7 percent for frontline office and administrative support (73,038 classified conversations), 58.1 percent for computer and mathematical occupations (336,486), 56.2 percent for food preparation and serving (2,948), 50.9 percent for personal care and service (4,439), 48.0 percent for sales, and 43.0 percent for all other occupations. The administrative-support estimate is based on a substantially larger cell than the other frontline estimates. The smaller frontline cells should therefore be interpreted with greater uncertainty.

The pattern is notable in light of experimental evidence. Shen and Tamkin (2026) find that AI assistance during skill acquisition reduced mastery, with the deficit concentrated among participants who delegated the task fully. Cognitive-engagement patterns preserved learning more effectively. The results here therefore suggest that frontline tasks receive not only less observed AI usage but also a mode composition associated with weaker human-capital accumulation. This comparison is descriptive: the AEI modes and the experimental interaction patterns are conceptually related but are not identical measures.

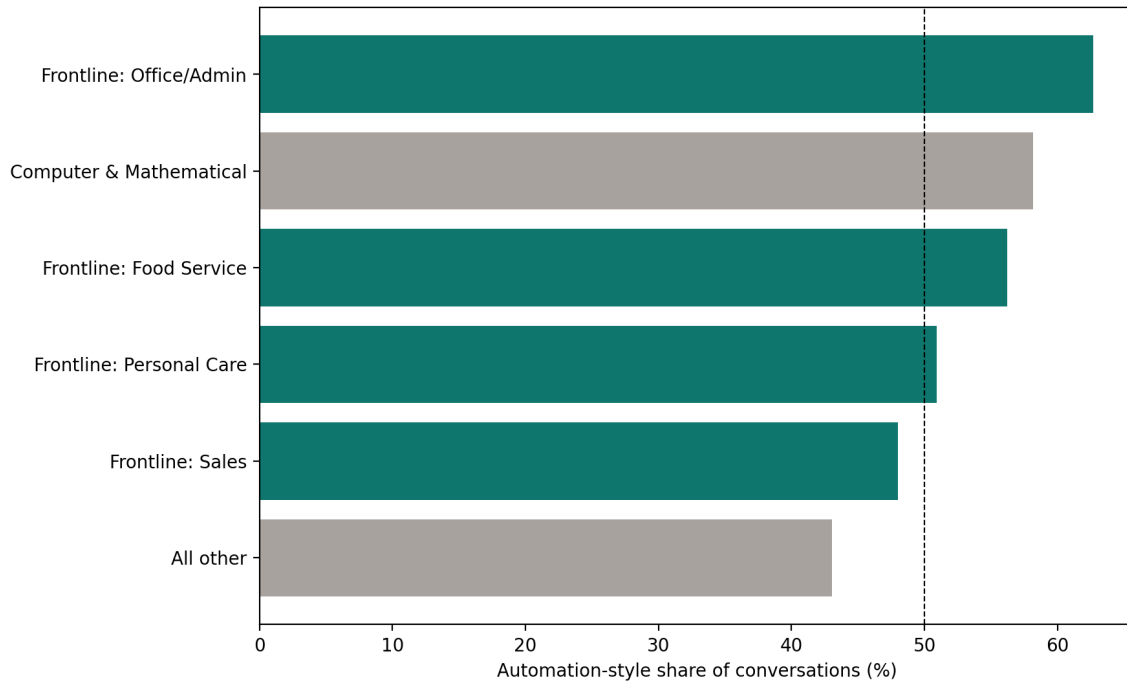


Figure 5. Automation-style share of classified conversations by occupation group.

Notes: Automation-style interactions combine directive and feedback-loop modes. The vertical reference line marks 50 percent. Teal bars denote frontline groups; gray bars denote comparison groups. Mode composition is conditional on observed use and may still reflect platform selection.

5.6 Observed Exposure, February 2026

The February 2026 cross-section reveals an important within-group pattern. Unweighted mean observed exposure is 0.38 for computer and mathematical occupations, 0.24 for sales, 0.19 for office and administrative support, 0.02 for personal care and service, and 0.01 for food preparation and serving. Forty-four of 112 frontline occupations, including all cook categories, waiters, and bartenders, register zero.

Within frontline work, exposure is sharply polarized. Customer Service Representatives score 0.70, above Software Developers at 0.29 and close to Computer Programmers at 0.75. Data Entry Keyers score 0.67. Within the same storefront, Retail Salespersons score 0.32, four times the Cashiers score of 0.08. The dividing characteristic is not occupational prestige but task medium: screen-mediated service tasks show high observed coverage, whereas physically co-present tasks show approximately none. This cross-section does not establish persistence on its own because the exposure measure is not commensurable with the 2025 usage shares. It does, however, reinforce the finding that embedded information tasks are the primary point of entry.

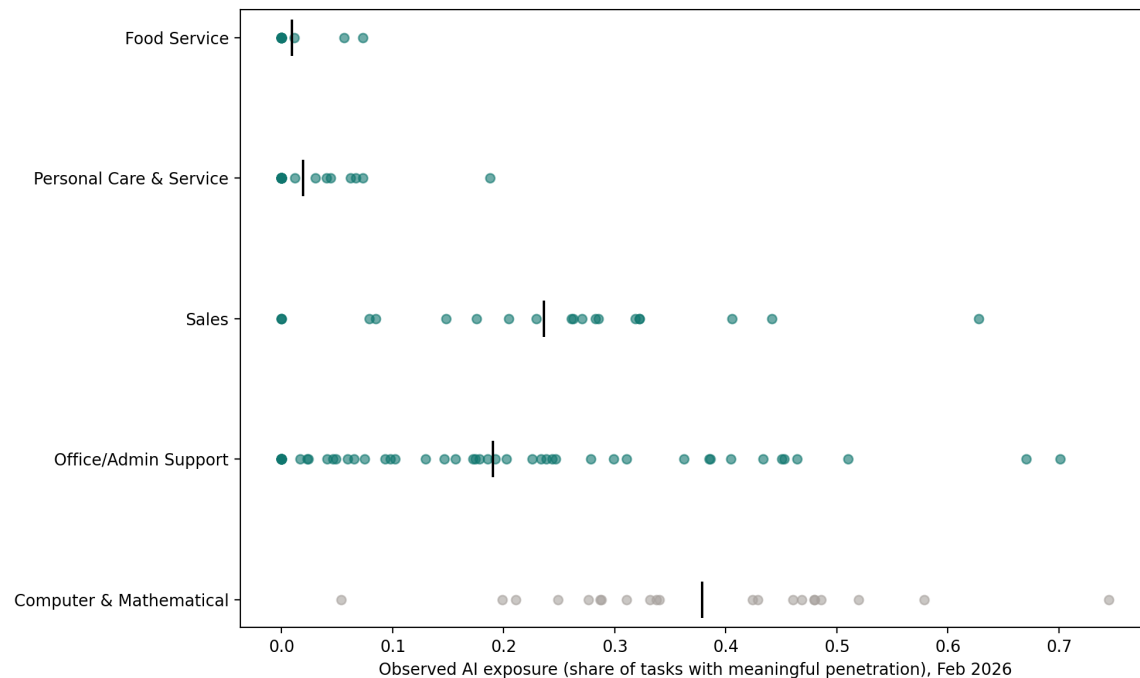


Figure 6. Observed AI exposure by occupation, February 2026.

Notes: Dots represent occupations and vertical bars represent unweighted group means. Teal denotes frontline occupation groups; gray denotes computer and mathematical occupations. Observed exposure is not directly comparable with the usage-share measures in Figures 1 and 4. Source: AEI labor-market-impacts release.

6. Robustness and Threats to Validity

Task splitting. Of 18,428 task statements, 272 (1.5 percent) appear under multiple occupations. The baseline assigns each shared task to one occupation. Splitting usage equally across all listing occupations leaves every frontline index unchanged to three decimal places and moves the computer and mathematical index from 10.944 to 10.918. The misclassified technical occupations identified in Section 5.2 hold their tasks uniquely, so exclusion rather than fractional assignment is the appropriate correction. The two data issues are independent.

Platform selection. Claude's user base is disproportionately technical, which mechanically raises computer and mathematical shares relative to an all-platform measure. Yin and Ogut (2026) quantify this directly: holding empirical design fixed and varying only the platform input changes their post-ChatGPT employment coefficient by a factor of 1.9, consumer- and enterprise-channel estimates within the same vendor disagree in sign, and reweighting to BLS workforce shares attenuates estimates by 42 to 93 percent. This is a larger correction than platform-selection critiques typically imply and should be taken seriously here. Three observations bound, without eliminating, its reach. First, within-frontline polarization occurs among occupations observed in the same platform environment, so user composition cannot generate it. Second, mode composition is estimated conditional on use, a distinct channel from the level bias Yin and Ogut quantify, although selection could still matter if selected users differ systematically in how they interact with AI. Third, the 2026 exposure scores incorporate first-party API traffic alongside consumer usage, a partial hedge against consumer-platform selection specifically. Given the magnitude Yin and Ogut report, the most defensible reading is that platform selection materially inflates the level of the headline representation gap, and a full correction using workforce-representative data, which this paper's data do not permit, would likely narrow it. The

within-group structure in Sections 5.2, 5.5, and 5.6 does not rely on cross-platform comparisons and is less directly implicated, although it cannot be fully separated from this bias with the data used here. Platform-independent Danish evidence also documents uneven occupational adoption using a wholly different method (Humlum and Vestergaard 2025), which is reassuring but not dispositive.

Weighting. The means reported in Section 5.6 weight occupations equally. Because several high-exposure frontline occupations also have large employment counts, employment weighting would likely raise the frontline means. The replication code is designed to report both unweighted and employment-weighted estimates when a detailed OEWS file is supplied.

Remaining threats. Conversation-to-task classification error is not directly observed. The 100 percent statement-level match rate validates the join but not the upstream classifier. Because occupation is inferred from task content, managerial performance of frontline-classified tasks cannot be excluded. Employer-deployed AI is also incompletely observed in consumer usage data. This omission leaves open the possibility that frontline workers encounter AI primarily through employer-deployed systems, but the current data cannot evaluate that mechanism directly.

7. Mechanisms

Three mechanisms could generate the observed configuration: a large per-worker usage deficit, continued underrepresentation across the 2025 releases, an automation tilt in interaction modes, and polarization by task medium.

Demand. The in-person core of frontline work (stocking, serving, de-escalating, and caring) cannot be performed easily through a language-model interface. This is consistent with the characterization of AI as reducing the cost of prediction and information processing rather than physical execution (Agrawal, Gans, and Goldfarb 2018), which implies adoption should concentrate in information-processing tasks rather than embodied service work; the task-medium boundary in Section 5.6 is the empirical counterpart. This mechanism explains much of the near-zero coverage in food service and personal care and the contrast between screen-mediated and physically co-present work. It cannot by itself explain low per-worker usage in occupations that contain many feasible information tasks.

Access. Frontline work is performed under workplace time, device, and policy constraints. In my experience operating Krishna Foods and Energy, a food manufacturing and retail business, the first processes to be digitized were demand forecasting, procurement comparisons, pricing, inventory reporting, and customer communication. Production-floor and retail tasks were harder to support because workers often relied on shared devices, had limited uninterrupted time, and needed to coordinate in noisy, physical environments. Even where an AI tool could assist with documentation or troubleshooting, opening a separate interface could add friction during a live operational problem. At the same time, supervisors and back-office staff could use the same tools more readily, which cautions against treating access as a uniform worker-level constraint. Low observed adoption may therefore reflect not only device and time constraints but also the absence of the complementary organizational investments, such as redesigned workflows and worker-facing deployment, that are required to convert feasible tasks into realized use (Agrawal, Gans, and Goldfarb 2018). The one setting with experimental evidence of large frontline-adjacent gains involved precisely such an investment: an employer-provisioned assistant integrated into the existing workflow (Brynjolfsson, Li, and Raymond 2025).

Measurement. Consumer usage data omit part of employer-embedded deployment. The direction of this omission is ambiguous for the overall level of frontline adoption, but the taxonomy audit shows that

correcting the measurable component of occupational misclassification enlarges rather than reduces the gap. Future work using employer-level deployment data is needed to determine whether unobserved systems substitute for or complement worker-directed tools.

On balance, the evidence suggests that demand and access jointly shape the level of the gap. Its composition (an entry point through embedded information tasks, stronger coverage of screen-mediated occupations, and an automation tilt conditional on use) appears across several related measurement systems, although platform selection and employer deployment remain important qualifications.

8. Conclusion and Policy Implications

Frontline service occupations account for roughly one-third of U.S. employment and one-ninth of observed AI usage. The taxonomy audit substantially lowers the apparent representation of administrative-support work, while task-splitting leaves the estimates essentially unchanged. The comparison of the February and August 2025 releases provides suggestive evidence of continued underrepresentation, and conditional on observed use, frontline tasks are more likely to appear in automation-style interactions. By February 2026, observed exposure was polarized within frontline work along the screen-mediated versus physically co-present task boundary, with customer-service occupations among the most exposed and many in-person service occupations at zero.

Two cautious implications follow. First, if productivity gains accrue primarily where adoption occurs, current patterns may direct gains toward occupations already intensive in AI usage while leaving a large share of the workforce with limited worker-directed access. Distributional analyses centered on professional occupations may therefore describe only the most exposed segment of the labor market. Second, interaction mode is plausibly a deployment choice rather than a fixed technological property. The strongest frontline-adjacent productivity evidence involves employer-provided augmentation (Brynjolfsson, Li, and Raymond 2025), while the learning evidence in Shen and Tamkin (2026) indicates that full delegation can carry human-capital costs. Product and policy choices about interface design, training, and worker control may therefore affect both productivity and skill formation.

Future research follows directly. A methodologically consistent adoption panel should reclassify a fixed task universe across AEI releases to convert the persistence comparison into a measured trajectory. Whether the June 2026 release (Cadences) supplies the monthly data this requires is not yet confirmed: its report emphasizes hourly and daily usage patterns and survey data rather than occupational task shares, and its data files are not yet in the public repository, so this extension should be checked against the actual files, if and when published, rather than treated as already in hand. A two-part occupation model should distinguish the probability of any observed usage from usage intensity conditional on entry. A supervisory decomposition that crosses task assignments with first-line-supervisor SOC codes could test whether apparent frontline adoption is, in practice, managerial adoption. Each extension is implementable with public occupational data and more complete release-level usage files.

Data and Code Availability

Replication code, processed outputs, and documentation are maintained at github.com/raamnandhakumar-eng/polecoai. The underlying Anthropic Economic Index, BLS OEWS, and O*NET data remain subject to their respective release terms.

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